Notes: Sapienza (JFE, 2004)

The Effects of Government Ownership on Bank Lending

Contents

1 Big picture

- **Question.** Does government ownership change bank lending behavior, and if so, does the evidence look more like social lending, bureaucratic agency problems, or political patronage?
- **Setting.** The paper studies Italy in 1991–1995, a period in which state-owned banks were economically important and politically connected appointments to bank leadership were observable.
- **Core data advantage.** The author uses individual credit-line contracts, not aggregate bank profitability. This allows comparison of loan rates charged by state-owned and private banks to the same or very similar firms.
- **Main empirical result.** State-owned banks charge systematically lower interest rates than private banks to comparable firms. The baseline difference is about 23 basis points in raw matched comparisons and about 44–50 basis points in controlled regressions.
- **Social-view test.** If state banks solve market failures, they should favor credit-constrained or small firms. The evidence is weak for this: the discount remains for firms that also borrow from private banks and have unused private credit lines.
- **Political-view evidence.** State-bank discounts are larger in southern Italy and larger for big firms. The strongest evidence is that discounts are larger in provinces where the political party affiliated with the state bank is electorally stronger.
- **Identification idea.** The strongest specifications compare rates for the same firm, at the same time, from different types of banks, using firm fixed effects and time fixed effects.
- **Economic takeaway.** State ownership of banks can distort credit allocation not merely by making banks inefficient, but by turning lending terms into a vehicle for political patronage.

2 Data and measurement

2.1 Institutional setting: Italian state-owned banks

- The Italian banking sector contained many private banks but state-owned banks held a large share of assets.
- In 1995, state-owned banks held roughly 58% of total bank assets in Italy.
- State-owned banks had formal and informal political links: before the early 1990s, chairpersons and top executives of many state banks were appointed through politically mediated procedures.
- Political appointments are crucial for the paper's final test because they allow the author to match a state bank to a political party and then to local vote shares.

Interpretation: The institutional setting provides unusually direct variation in political influence over banks. The paper is not simply comparing public and private banks; it also asks whether the public banks behave differently where their affiliated politicians are stronger.

2.2 Loan and firm data

- The two main databases come from *Centrale dei Bilanci*: the Company Accounts Dataset and the Credit Register.
- The Company Accounts Dataset contains balance sheets and income statements for more than 50,000 Italian firms.
- The Credit Register contains individual loan contracts above 80 million lire.

- Ninety banks, covering more than 80% of total lending, report detailed loan interest rates.
- The author focuses on credit lines, excluding long-term, collateralized, and subsidized loans to keep loan types more homogeneous.
- The final matched sample contains 6,968 companies and 110,786 company–bank–year observations: 55,393 involving state-owned banks and 55,393 involving private banks.

Interpretation: Credit lines are a useful object because they are common, repeatable, and relatively standardized. This reduces the risk that observed rate differences arise from comparing completely different loan products.

2.3 Risk scores and matching

- Firms are assigned a numerical risk score from the CdB Diagnostic System.
- The score is based on discriminant analysis and is meant to summarize a firm's probability of distress.
- Banks in the CdB system had access to the same risk information before lending.
- When a firm borrows from both state-owned and private banks, the firm can be matched to itself.
- When no self-match is available, the paper matches firms by year, industry, geography, ownership type, risk profile, and similar sales.

Interpretation: Matching on risk score and firm characteristics is essential. If state banks lent to safer firms, lower rates would be unsurprising. The matching and fixed effects are designed to rule out that simple selection story.

2.4 Interest rate measure

The dependent variable is the relative interest rate:

$$r_{ikt} = \text{Rate}_{ikt} - \text{Prime}_t, \tag{1}$$

where Rate_{ikt} is the interest rate charged to firm i by bank k in year t, and Prime_t is the prime rate.

The loan rate is measured as

$$\text{Rate}_{ikt} = \frac{\text{quarterly interest payments plus fixed fees}}{\text{quarterly average loan balance}}. \tag{2}$$

Interpretation: Subtracting the prime rate removes common interest-rate movements. The remaining spread is the price component plausibly affected by bank ownership, firm risk, market power, and political incentives.

3 Theoretical views and predictions

3.1 Social view

- State-owned banks exist because private markets fail to allocate credit efficiently.
- They should channel funds toward socially valuable firms, credit-constrained borrowers, small firms, or depressed regions.
- Lower interest rates from state banks could be efficient if they correct market failures.

Interpretation: The social view predicts targeted subsidies to constrained or socially valuable borrowers, not broad discounts to firms that can already borrow privately.

3.2 Agency view

- Governments may have social objectives, but state-bank managers face weak incentives.
- Low-powered incentives can generate misallocation, slack, or favoritism unrelated to social welfare.
- State banks might favor large firms because managers get personal or career benefits from relationships with large borrowers.

Interpretation: The agency view can explain inefficiency, but it does not naturally predict that lending discounts increase in provinces where the bank's affiliated political party is stronger.

3.3 Political view

- Politicians use state-owned enterprises to maintain political support.
- State-owned banks can transfer resources through below-market loan rates.
- Discounts should be larger where patronage is more valuable: politically important regions, large employers, and locations where the affiliated party is strong.

Interpretation: The political view is the sharpest explanation for the paper's final evidence: a party-affiliated state bank grants larger discounts in areas where that party receives more votes.

4 Empirical specifications

4.1 Baseline ownership regression

The baseline specification is

$$r_{ikt} = \alpha_i + \tau_t + \beta \mathrm{STATE}_{kt} + \Gamma' X_{kt} + \Lambda' Z_{it} + \varepsilon_{ikt}, \tag{3}$$

where

- α_i are firm fixed effects;
- τ_t are time fixed effects;
- STATE_{kt} equals one if bank k is state-owned at time t;
- X_{kt} includes bank size and nonperforming loan share;
- Z_{it} includes firm size and firm risk score;
- some specifications include province-level bank concentration, HHI, and $\mathrm{STATE}_{kt} \times \mathrm{HHI}$.

Interpretation: A negative β means state-owned banks charge lower spreads than private banks to the same firm in the same year, after controlling for bank, firm, and market characteristics.

4.2 Credit-constraint test

To test whether state banks lend to constrained firms, the paper restricts attention to firms that borrow from both types of banks and then adds interactions with credit-line usage at private banks:

$$r_{ikt} = \alpha_i + \tau_t + \beta \text{STATE}_{kt} + \sum_q \theta_q \left(\text{STATE}_{kt} \times HighUsage^q_{it}\right) + Controls + \varepsilon_{ikt}. \tag{4}$$

Interpretation: If state banks were mainly lending to credit-constrained firms, the discount should be larger when firms heavily use private credit lines. The interaction tests show that this is not the main explanation.

4.3 Regional heterogeneity

For geographical tests, the paper estimates

$$r_{ikt} = \alpha_i + \tau_t + \beta \text{STATE}_{kt} + \delta_S(\text{STATE}_{kt} \times South_i) + \delta_N(\text{STATE}_{kt} \times North_i) + Controls + \varepsilon_{ikt}, \tag{5}$$

where the center is effectively the omitted location in the interaction structure used in the table.

Interpretation: Larger discounts in the south can be consistent with either social subsidies to depressed areas or political patronage. This test alone cannot fully distinguish those views, but it is hard to explain using a simple managerial-agency story.

4.4 Firm-size heterogeneity

For firm size, the paper estimates

$$r_{ikt} = \alpha_i + \tau_t + \beta \text{STATE}_{kt} + \sum_{q=1}^{4} \delta_q(\text{STATE}_{kt} \times SizeQuintile_{iq}) + Controls + \varepsilon_{ikt}, \tag{6}$$

where the largest size quintile is the omitted group.

Interpretation: If social lending helps small constrained firms, the smallest firms should get the largest discount. Instead, the largest firms receive the largest discount, which points away from the social view.

4.5 Political-affiliation specification

For state-owned banks only, the key political regression is

$$r_{ikt} = \alpha_i + \tau_t + \phi LocalStrength_{party(k),province(i),t} + Controls + \varepsilon_{ikt}. \tag{7}$$

- $LocalStrength$ is the vote share of the political party affiliated with the chairperson of state-owned bank k in the province where firm i borrows.
- The author identifies political affiliations for 36 state-owned banks.
- Electoral results come from the 1987, 1992, and 1994 national elections at the provincial level.

Interpretation: A negative ϕ means that affiliated state banks charge lower rates in areas where their affiliated party is electorally stronger. This is the paper's most direct evidence of patronage.

5 Identification and empirical design

5.1 What identifies the ownership effect

- The paper compares state and private bank loan rates for firms with the same observable risk profile.
- Firm fixed effects allow comparisons across banks lending to the same firm.
- Time fixed effects absorb common credit-market conditions.
- Bank controls address size, risk, and market concentration.
- The dual-borrower sample checks whether the result survives when firms can access both state and private credit.

Interpretation: The central identifying assumption is that, after firm fixed effects and controls, remaining differences in loan rates reflect bank objectives rather than unobserved loan quality or borrower risk.

5.2 Why the political test is especially informative

- A generic state-bank discount could be social, agency-driven, or political.
- A discount that varies with the local vote share of the affiliated political party is much more difficult to explain with the social or agency views alone.
- The political test uses variation across provinces and parties within the state-owned bank sample.

Interpretation: The political specification moves the paper from showing that state banks lend cheaply to showing that their cheap lending is aligned with political incentives.

5.3 Limits and caveats

- The paper cannot observe every dimension of loan quality, service quality, or revocation probability.
- The sample is not the entire loan portfolio of each bank.
- Political affiliation is measured through chairpersons and newspaper sources; it is informative but not perfect.
- The setting is Italy in the early 1990s, so generalization requires caution.

Interpretation: The paper is strongest as evidence that politicized credit allocation can exist and be measured, not as a universal estimate of the welfare cost of state banking in all countries.

6 Tables and figures

The paper has no standalone figures. The key visual material is the sequence of original tables, cropped directly from the paper and grouped compactly where possible.

6.1 Tables 1–2: Bank and borrower samples

Read:

- Table 1 shows that state-owned banks are larger on average than private banks: mean total assets are 27,314 billion lire for state-owned banks versus 11,856 billion lire for private banks.
- State-owned banks have higher nonperforming loans: 8.41% of loans versus 6.14% for private banks.
- State-owned banks have lower return on assets: 0.28% versus 0.46%.
- Table 2 shows that the matched firm samples are extremely similar: both private-bank and state-bank borrowers have median assets of 18 billion lire, median sales of 21 billion lire, median employment of 58, and median leverage around 71%.

Economic message: The bank comparison shows why simple raw differences could be confounded by bank characteristics. The firm comparison shows that the matching procedure creates a credible borrower-level comparison group.

Additional interpretation: The most important point is the asymmetry between the bank and firm panels. Banks differ meaningfully in size, riskiness, and profitability; firms are deliberately matched to look nearly identical. That is why later regressions must control for bank-level characteristics but can use firm fixed effects to absorb borrower quality.

Table 1
Summary statistics: the bank sample
Panel A shows summary statistics for the sample of privately owned banks (bank–years). Panel B shows summary statistics for state-owned banks. Return on assets is earnings over total assets. Operating costs include wages and other operating costs.

Variable	Mean	Median	Std dev.	Min	Max	Obs.
Panel A: Privately owned banks						
Total assets (bill. of lire)	11,856	6,318	16,674	186	110,531	192
Total loans (bill. of lire)	4,917	2,570	6,604	80	44,820	192
Percentage of loans over total assets	42.90	43.00	5.14	30.00	56.00	192
Percentage of nonperforming loans to total loans	6.14	5.25	4.02	1.42	23.63	192
Return on assets (%)	0.46	0.51	0.62	−6.47	1.28	192
Operating costs over total assets (%)	3.04	2.87	0.78	1.56	5.96	192
Panel B: State-owned banks						
Total assets (bill. of lire)	27,314	6,070	40,192	547	188,944	199
Total loans (bill. of lire)	12,292	2,586	18,750	218	79,011	199
Percentage of loans over total assets	41.75	42.00	7.59	24.00	70.00	199
Percentage of nonperforming loans to total loans	8.41	6.91	6.23	1.63	39.55	199
Return on assets (%)	0.28	0.34	0.73	−7.19	1.32	199
Operating costs over total assets (%)	3.05	3.05	0.65	1.73	5.65	199

(a) Table 1: bank summary statistics.

Table 2
Summary statistics: the company sample
Summary statistics for the two subsamples of company–bank–years. Panel A shows the summary statistics for the subsample of companies that borrow from privately owned banks (company–bank–year). Panel B shows the summary statistics for the companies that borrow from state-owned banks. Total Assets is beginning-of-year total assets in lire. Sales is beginning-of-year sales in lire. Employees is the number of employees at the beginning of the year. Return on sales is earning before interest, taxes, and depreciation (*EBITDA*) over sales. Age is the number of years since incorporation. Leverage is book value of short- plus long-term debt divided by the sum of book value of short- plus long-term debt and book value of equity. Coverage is interest expense divided by *EBITDA* (I truncate values above 100 at 100 and values below zero at zero).

Variable	Mean	Median	Std dev.	Obs.
Panel A: Companies borrowing from privately owned banks				
Total assets (bill.)	101	18	546	55,393
Sales (bill.)	108	21	654	55,393
Employees	231	58	928	54,782
Return on sales	8.48	7.98	7.25	54,799
Age	25	18	36	55,168
Leverage	68.11	70.69	17.59	54,638
Coverage	1.85	1.47	2.56	55,351
Panel B: Companies borrowing from state-owned banks				
Total assets (bill.)	101	18	547	55,393
Sales (bill.)	108	21	654	55,393
Employees	231	58	928	54,789
Return on sales	8.51	7.95	7.37	54,817
Age	25	18	36	55,173
Leverage	68.12	70.71	17.60	54,646
Coverage	1.86	1.47	2.63	55,349

(b) Table 2: firm summary statistics.

6.2 Table 3: Interest rates by risk category

Read:

- State-owned banks charge lower rates in every risk category.

- Across all borrowers, state-owned banks charge 3.07% above the prime rate while private banks charge 3.31%, a raw difference of -23 basis points.
- Among firms that borrow from both state-owned and private banks, the all-borrower difference remains -23 basis points.
- The vulnerable category has the largest visible discount: about -41 basis points.

Economic message: The lower rate is not caused by one outlier risk group or by a comparison between totally different firms. Even firms that borrow from both types of banks receive cheaper state-bank credit.

Additional interpretation: This table is the first blow against a simple adverse-selection story. If state banks merely served safer borrowers, the discount would vanish once firms are grouped by risk. It does not. The persistence of the discount in Panel B is particularly important because the same firm can be compared across different bank ownership types.

Table 3
Interest rates charged by state-owned and privately owned banks by loan risk category
I define the interest rate paid by the firm to the bank as the ratio of the quarterly payment (interest plus fees) to its quarterly average balance minus the prime rate. The loan risk category is based on the numerical score of the company (see details in Appendix A). Difference is the average difference between the second column (interest rates charged by state-owned banks) and the third column (interest rates charged by privately owned banks). I test the statistical significance of the difference using the *t*-statistic with reference to a mean of zero. ***, ** indicate statistically significant at the 1% and 5% level, respectively.

Risk category	State-owned banks	Privately owned banks	Difference	Obs.
Panel A: Whole sample				
Highly secure	2.53	2.75	−0.22***	1,420
Secure	2.75	2.97	−0.22***	15,262
Vulnerable	2.84	3.25	−0.41***	409
Highly vulnerable	3.05	3.28	−0.24***	11,743
Uncertainty between vulnerability and risk	3.18	3.43	−0.25***	13,471
Risk of bankruptcy	3.36	3.58	−0.22***	10,472
High risk of bankruptcy	3.69	3.80	−0.11**	2,616
All borrowers	3.07	3.31	−0.23***	55,393
Panel B: Firms borrowing from both state-owned and privately owned banks				
Highly secure	2.52	2.74	−0.22***	1,360
Secure	2.72	2.94	−0.22***	13,373
Vulnerable	2.85	3.26	−0.41***	394
Highly vulnerable	3.02	3.27	−0.24***	10,248
Uncertainty between vulnerability and risk	3.15	3.40	−0.25***	11,899
Risk of bankruptcy	3.33	3.54	−0.21***	9,184
High risk of bankruptcy	3.66	3.79	−0.13**	2,438
All borrowers	3.05	3.27	−0.23***	48,896

Figure 2: Table 3: interest rates by risk category.

6.3 Table 4: Baseline ownership regressions

Read:

- Panel A shows that the basic state-bank coefficient is -0.2378 with firm and time fixed effects, close to the raw -23 basis point result in Table 3.

- After adding bank size, nonperforming loans, HHI, interactions, firm size, and firm score, the state-bank coefficient is around -0.44 to -0.50.
- Panel B restricts to firms that borrow from both state and private banks. The state-bank discount remains about -0.44 to -0.50.
- Panel C adds private-credit-line usage interactions. The state-bank discount remains around -0.44, and the usage interactions are not economically central.

Economic message: The controlled regressions increase rather than eliminate the state-bank discount. The discount is not explained by state banks being smaller, safer, more concentrated, or serving credit-constrained borrowers.

Additional interpretation: Table 4 is the core empirical table. Its logic is sequential: first show the discount; then add bank controls; then restrict to dual borrowers; then check credit-line usage. The estimate remains stable across these increasingly demanding comparisons, which makes it difficult to interpret the discount as a simple borrower-selection artifact.

Table 4
Interest rates charged by state-owned and privately owned banks
The dependent variable is the interest rate charged to firm *i* by bank *k* at time *t* minus the prime rate at time *t*. $STATE_{k,t}$ is a dummy variable equal to one if at time *t* bank *k* is a state-owned bank. I measure the size of the bank by logarithm of total assets. The percentage of nonperforming loans is the ratio of non-performing loans to total loans. I measure market concentration at the province level by the Herfindahl-Hirschman Index (HHI) on total banking lending. The size of the firm is the logarithm of sales. All regressions include year and firm dummies. Heteroskedasticity-robust standard errors are in brackets. The standard errors are corrected for within-year clustering. ***, ** indicate statistically significant at the 1% and 5% level, respectively. The table also reports the *p*-value of an *F*-test for the hypothesis that the joint effect of all the variables equals zero. Panel A reports the results for the whole sample. Panels B and C report the results for the subsample of firms that borrow from both state-owned and privately owned banks, respectively.

	(1)	(2)	(3)	(4)	(5)
Panel A					
$State_{k,t}$	−0.2378***	−0.4589***	−0.5019***	−0.4417***	−0.4424***
	(0.0274)	(0.0166)	(0.0180)	(0.0218)	(0.0218)
Size of the bank		0.1936***	0.1730***	0.1723***	0.1728***
		(0.0078)	(0.0037)	(0.0041)	(0.0040)
Percentage of nonperforming loans			0.0337***	0.0338***	0.0336***
			(0.0014)	(0.0014)	(0.0015)
Concentration of loans (HHI)			2.6681***	3.0753***	2.8267***
			(0.4417)	(0.3514)	(0.4197)
Concentration of loans if $State_{k,t} = 1$				−0.8677***	−0.8561***
				(0.3223)	(0.3183)
Size of the firm					−0.2453***
					(0.0051)
Score of the firm					0.0365***
					(0.0081)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes	Yes
Observations	110,786	110,786	110,752	110,752	110,752
Adjusted *R*-squared	0.407	0.420	0.425	0.425	0.428
p-Value of *F*-test for total effect equal to zero	0.0000	0.0000	0.0000	0.0000	0.0000
Panel B					
$State_{k,t}$	−0.2293***	−0.4510***	−0.4980***	−0.4374***	−0.4376***
	(0.0261)	(0.0185)	(0.0168)	(0.0221)	(0.0220)
Size of the bank		0.1895***	0.1694***	0.1689***	0.1691***
		(0.0087)	(0.0042)	(0.0046)	(0.0045)
Percentage of nonperforming loans			0.0341***	0.0343***	0.0340***
			(0.0014)	(0.0014)	(0.0014)
Concentration of loans (HHI)			2.8282***	3.2309***	2.9066***
			(0.4835)	(0.4297)	(0.4958)
Concentration of loans if $State_{k,t} = 1$				−0.8756***	−0.8685***
				(0.3341)	(0.3322)
Size of the firm					−0.2648***
					(0.0065)
Score of the firm					0.0335***
					(0.0077)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes

Table 4. (*Continued*)

	(1)	(2)	(3)	(4)	(5)
Time fixed effect	Yes	Yes	Yes	Yes	Yes
Observations	97,792	97,792	97,760	97,760	97,760
Adjusted *R*-squared	0.407	0.420	0.425	0.423	0.427
p-Value of *F*-test for total effect equal to zero	0.0000	0.0000	0.0000	0.0000	0.0000

Panel C

	(1)	(2)	(3)
$State_{k,t}$	−0.4402***	−0.4382***	−0.4373***
	(0.0220)	(0.0225)	(0.0220)
Size of the bank	0.1690***	0.1691***	0.1691***
	(0.0045)	(0.0045)	(0.0045)
Percentage of nonperforming loans	0.0340***	0.0340***	0.0340***
	(0.0014)	(0.0014)	(0.0014)
Concentration of loans (HHI)	2.9151***	2.9104***	2.9071***
	(0.4914)	(0.4976)	(0.4946)
Concentration of loans if the bank is state-owned	−0.8838***	−0.8751***	−0.8672***
	(0.3284)	(0.3258)	(0.3323)
Size of the firm	−0.2646***	−0.2647***	−0.2649***
	(0.0065)	(0.0066)	(0.0064)
Score of the firm	0.0335***	0.0335***	0.0334***
	(0.0076)	(0.0076)	(0.0077)
$State_{k,t} = 1$ if the firm has more than 8% of credit line usage	0.0150		
	(0.0127)		
$State_{k,t} = 1$ if the firm has more than 15% of credit line usage		0.0214	
		(0.0303)	
$State_{k,t} = 1$ if the firm has more than 37% of credit line usage			−0.0364
			(0.0456)
Firm fixed effect	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes
Observations	97760	97760	97760
Adjusted *R*-squared	0.427	0.427	0.427
p-value of *F*-test for total effect equal to zero	0.0000	0.0000	0.0000

6.4 Tables 5–6: Regional heterogeneity

Read:

- Table 5 shows larger raw discounts outside the north: -18 basis points in the north, -39 basis points in the center, and -45 basis points in the south.

- Table 6 confirms this in regressions. The baseline state-bank discount is about -45 basis points in the preferred controlled specifications.
- The additional state-bank discount for southern firms is about -31 basis points in the full specification.
- The northern interaction is also negative, but the south effect is larger, consistent with especially favorable treatment in poorer or more patronage-intensive areas.

Economic message: Regional patterns are compatible with both a social subsidy interpretation and a political patronage interpretation. The south is poorer, but also historically more patronage-driven.

Additional interpretation: The regional results are a hinge point in the paper. They do not by themselves prove political lending because a benevolent government might want to help poorer southern firms. But they show that state-bank discounts are not uniform public-bank pricing mistakes; they are targeted in ways that align with political economy. This motivates the stronger electoral-affiliation test in Table 8.

Table 5
Differences in interest rates charged by state-owned and privately owned banks by region and borrower size
I define the interest rate as the ratio of the quarterly payment (interest plus fees) paid to the bank by the firm to its quarterly average balance, minus the prime rate. Difference is the average difference between the second column (interest rates charged by state-owned banks) and the third column (interest rates charged by privately owned banks). North includes the following regions: Piedmont, Valle d'Aosta, Lombardy, Trentino, Veneto, Friuli Venezia Giulia, Liguria, and Emilia Romagna. Center includes Tuscany, Umbria, Marche, and Lazio. South includes Abruzzo Molise, Campania, Puglia, Basilicata, Calabria, Sicily, and Sardinia. For the difference, I test statistical significance using the t-statistic with reference to a mean of zero. *** indicates statistically significant at the 1% level, ** indicates statistically significant at the 5% level.

Interest rate-prime:	State-owned banks	Privately owned banks	Difference	Obs.
Borrowers classified by geographical location				
North	3.03	3.22	−0.18***	38,786
Center	3.02	3.41	−0.39***	6,292
South	3.20	3.65	−0.45***	3,818
Borrowers classified by size				
First quintile in sales	3.70	3.86	−0.16***	9,780
Second quintile in sales	3.34	3.55	−0.21***	9,778
Third quintile in sales	3.12	3.36	−0.24***	9,780
Fourth quintile in sales	2.84	3.10	−0.25***	9,780
Fifth quintile in sales	2.23	2.51	−0.28***	9,778
All borrowers	3.05	3.27	−0.23***	48,896

(a) Table 5: raw differences by region and firm size.

Table 6
Regression results on interest rates charged by state-owned and privately owned banks in different areas
The dependent variable is the interest rate charged to firm i by bank k at time t minus the prime rate at time t. $STATE_{k,t}$ is a dummy variable equal to one if at time t bank k is a state-owned bank. North includes the following regions: Piedmont, Valle d'Aosta, Lombardy, Trentino, Veneto, Friuli Venezia Giulia, Liguria, and Emilia Romagna. Center includes Tuscany, Umbria, Marche, and Lazio. South includes Abruzzo Molise, Campania, Puglia, Basilicata, Calabria, Sicily, and Sardinia. I measure the size of the bank by logarithm of total assets. The percentage of nonperforming loans is the ratio of nonperforming loans to total loans. I measure market concentration at the province level by the Herfindahl-Hirschman Index (HHI) on total banking lending. Size of the firm is the logarithm of sales. All regressions include year and firm dummies. Heteroskedasticity-robust standard errors are in brackets. The standard errors are corrected for within-year clustering. ***, ** indicate statistically significant at the 1% and 5% level, respectively. The table also reports the p-value of an F-test for the hypothesis that the joint effect of all the variables equals zero.

	(1)	(2)	(3)	(4)	(5)
$State_{k,t}$	−0.1806***	−0.4102***	−0.4490***	−0.4562***	−0.4566**
	(0.0231)	(0.0195)	(0.0143)	(0.0203)	(0.0202)
State if firm is located in the South	−0.2709***	−0.2370***	−0.3097***	−0.3132***	−0.3143**
	(0.0401)	(0.0683)	(0.0365)	(0.0370)	(0.0365)
State if firm is located in the North	−0.2137***	−0.1501***	−0.1814***	−0.1818***	−0.1815***
	(0.0112)	(0.0123)	(0.0145)	(0.0154)	(0.0154)
Size of the bank		0.1870***	0.1655***	0.1655***	0.1657***
		(0.0097)	(0.0058)	(0.0057)	(0.0056)
Percentage of nonperforming loans			0.0356***	0.0356***	0.0353***
			(0.0016)	(0.0016)	(0.0016)
Concentration of loans (HHI)			2.7801***	2.7296***	2.4032***
			(0.4705)	(0.4473)	(0.5120)
Concentration of loans if $State_{k,t} = 1$				0.1093	0.1200
				(0.3012)	(0.2977)
Size of the firm					−0.2652***
					(0.0064)
Score of the firm					0.0334***
					(0.0076)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes	Yes
Observations	97,792	97,792	97,760	97,760	97,760
Adjusted R-squared	0.408	0.420	0.426	0.426	0.428
p-Value of F-test for total effect equal to zero	0.0000	0.0000	0.0000	0.0000	0.0000

(b) Table 6: regional interaction regressions.

6.5 Table 7: Firm-size heterogeneity

Read:

- Table 5's raw size panel shows nearly monotonic discounts by sales quintile: the largest firms receive the biggest discount.
- Table 7 confirms this pattern in regressions.
- The largest firms, which are the omitted category, receive an estimated state-bank discount around 48–57 basis points depending on specification.

- Smaller-firm interaction terms are positive, meaning their discount is smaller than the discount for the largest firms.

Economic message: The size evidence runs against the simple social view. If state banks primarily corrected credit constraints, smaller firms should be favored more heavily. Instead, large firms benefit more.

Additional interpretation: This result helps separate regional development from political patronage. Large firms are often politically valuable because they employ many workers, are visible, and can create concentrated electoral benefits. A subsidy to large borrowers is therefore consistent with political objectives, even if it is hard to justify as help for financially constrained firms.

Table 7
Regression results on interest rates charged by state-owned and privately owned banks by firm size
The dependent variable is the interest rate charged to firm i by bank k at time t minus the prime rate at time t. $STATE_{k,t}$ is a dummy variable equal to one if at time t bank k is a state-owned bank. I measure the size of the bank by logarithm of total assets. The percentage of nonperforming loans is the ratio of nonperforming loans to total loans. I measure market concentration at the province level by the Herfindahl-Hirschman Index (HHI) on total banking lending. Size of the firm is the logarithm of sales. All the regressions include year and firm dummies. Heteroskedasticity-robust standard errors are in brackets. The standard errors are corrected for within-year clustering. ***, ** indicate statistically significant at the 1% and 5% level, respectively. The table also reports the p-value of an F-test for the hypothesis that the joint effect of all the variables equals zero.

	(1)	(2)	(3)	(4)	(5)
$State_{k,t}$	−0.2965***	−0.5151***	−0.5703***	−0.4937***	−0.4792***
	(0.0301)	(0.0427)	(0.0324)	(0.0321)	(0.0323)
$State_{k,t}$ if firm in smallest size quintile	0.1906***	0.1868***	0.2006***	0.2113***	0.1582***
	(0.0366)	(0.0338)	(0.0380)	(0.0377)	(0.0391)
$State_{k,t}$ if firm in second size quintile	0.0933**	0.0899**	0.1038**	0.1104**	0.0859**
	(0.0419)	(0.0411)	(0.0443)	(0.0446)	(0.0417)
$State_{k,t}$ if firm in third size quintile	0.0466	0.0396	0.0483	0.0536	0.0359
	(0.0431)	(0.0398)	(0.0412)	(0.0413)	(0.0401)
$State_{k,t}$ if firm in fourth size quintile	0.0287	0.0272	0.0320	0.0356	0.0258
	(0.0253)	(0.0231)	(0.0241)	(0.0242)	(0.0237)
Size of the bank		0.1894***	0.1692***	0.1684***	0.1688***
		(0.0085)	(0.0040)	(0.0044)	(0.0043)
Percentage of nonperforming loans			0.0344***	0.0346***	0.0342***
			(0.0013)	(0.0013)	(0.0013)
Concentration of loans (HHI)			2.8206***	3.3630***	3.0251***
			(0.4919)	(0.4608)	(0.5301)
Concentration of loans if $State_{k,t} = 1$				−1.1806***	−1.0980***
				(0.3375)	(0.3233)
Size of the firm					−0.2476***
					(0.0078)
Score of the firm					0.0334***
					(0.0076)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes	Yes
Observations	97,792	97,792	97,760	97,760	97,760
Adjusted R-squared	0.407	0.420	0.425	0.425	0.427

Table 8
Regression results on interest rates charged by state-owned banks by electoral results, party affiliation and
The dependent variable is the interest rate charged to firm i by bank k at time t minus the prime rate at time t. I measure the local political strength of the party by the percentage of votes received by the party to which the chairperson of the state-owned bank is affiliated in the area where the firm is borrowing. I measure the size of the bank by logarithm of total assets. The percentage of nonperforming loans is the ratio of nonperforming loans over total loans. I measure market concentration at the province level by the Herfindahl-Hirschman Index (HHI) on total banking lending. Size of the firm is the logarithm of sales. All the regressions include year and firm dummies. Heteroskedasticity-robust standard errors are in brackets. The standard errors are corrected for within-year clustering. ***, ** indicate statistically significant at the 1% and 5% level, respectively. The table also reports the p-value of an F-test for the hypothesis that the joint effect of all the variables equals zero. The Column 1 sample includes all observations (state-owned-banks–firm–year) for which the political affiliation of the chairperson of the state-owned bank is available. Column 2 is the same sample, excluding 1994 and 1995. Columns 3 and 4 include only loans from national state-owned banks.

	(1)	(2)	(3)	(4)
Local political strength of the party	−0.2001**	−0.2295***	−0.3240***	−0.2837***
	(0.0806)	(0.0844)	(0.1239)	(0.1005)
Size of the bank	0.1702***	0.1641***	0.1282***	
	(0.0065)	(0.0033)	(0.0237)	
Percentage of nonperforming loans	0.0303***	0.0230***	0.0206***	
	(0.0045)	(0.0030)	(0.0061)	
Concentration of loans (HHI)	7.3368***	8.0807***	7.9113***	7.7236***
	(1.2113)	(0.7930)	(0.7298)	(0.7004)
Size of the firm	−0.3744***	−0.3277***	−0.3432***	−0.3435***
	(0.0742)	(0.0690)	(0.0853)	(0.0849)
Score of the firm	0.0321***	0.0328**	0.0258	0.0258
	(0.0087)	(0.0144)	(0.0167)	(0.0167)
Time fixed effect	Yes	Yes	Yes	Yes
Firm fixed effect	Yes	Yes	Yes	Yes
Bank fixed effect	No	No	No	Yes
Observations	26,698	25,049	17,671	17,671
Adjusted R-squared	0.4881	0.4953	0.5088	0.5087

6.6 Table 8: Electoral strength and political affiliation

Read:

- Table 8 uses only state-owned-bank loans and asks whether rates vary with local political strength of the party affiliated with the state bank.
- The coefficient on local political strength is negative in all columns.
- The estimated coefficient ranges from about -0.20 to -0.32 depending on controls and sample.
- The effect remains statistically significant after controlling for bank size, nonperforming loans, HHI, firm size, firm score, fixed effects, and alternative sample restrictions.

Economic message: Borrowers pay lower rates when the party linked to the state bank is stronger in their province. This is the paper's most direct evidence that state-bank lending terms are tied to political incentives.

Additional interpretation: Table 8 is what makes the political interpretation dominant. The earlier tables show state banks lend cheaply; Table 8 shows that cheap lending is geographically aligned with the political strength of bank-affiliated parties. A pure agency story would predict slack or mispricing, but not this local vote-share pattern. A pure social story would require the political party vote share to proxy exactly for social need, which is much less convincing.

6.7 Tables 9–10: Appendix risk-score and electoral data

Read:

- Table 9 shows the risk-score distribution. By construction, the matched state-bank and private-bank subsamples have the same risk profile.
- The largest categories are secure, uncertainty between vulnerability and risk, highly vulnerable, and risk of bankruptcy.
- Table 10 summarizes party vote shares in provinces where affiliated state banks lent.
- Christian Democrats and Socialist Party affiliates are the main political groups in the 1987 and 1992 panels; after the 1994 political transition, the Socialist Party and National Alliance appear in the relevant sample.

Economic message: These appendix tables document two foundations of the empirical design: matched borrower risk and measurable local political strength.

Additional interpretation: Table 9 gives credibility to the loan-pricing comparison because borrower risk is balanced by construction. Table 10 gives credibility to the political test because party strength varies meaningfully across provinces. Together, the appendix tables show that the paper's core empirical variables are not vague institutional anecdotes; they are operationalized in data.

Table 9

Score	Frequency	Percent	Cumulative Frequency
High secure	1,420	2.6	1,420
Secure	15,262	27.6	16,682
Vulnerable	409	0.7	17,091
Highly vulnerable	11,743	21.2	28,834
Uncertainty between vulnerability and risk	13,471	24.3	42,305
Risk of bankruptcy	10,472	18.9	52,777
High risk of bankruptcy	2,616	4.7	55,393

(a) Table 9: risk-score distribution.

Table 10

	Mean	Std. dev.	Min	Max	Number of provinces
Panel A: 1987 election					
Christian Democrats	0.34565932	0.08607512	0.08375919	0.52429986	80
Socialist Party	0.13883011	0.028753	0.06012196	0.20933744	89
Italian Liberal Party	0.02274278	0.01556027	0.00539812	0.09755591	52
Social Democratic Party	0.02849019	0.01601734	0.00475035	0.09887846	71
Panel B: 1992 election					
Christian Democrats	0.30600768	0.09383184	0.07377624	0.51638401	89
Socialist Party	0.13649238	0.03488646	0.04451371	0.26654419	91
Italian Liberal Party	0.02806362	0.01906862	0.01006228	0.13501064	70
Social Democratic Party	0.02611314	0.01770669	0.00560963	0.07984234	76
Panel C: 1994 election					
Socialist Party	0.0183175	0.0066557	0.0094487	0.0324773	33
National Alliance	0.1293044	0.0635202	0.0404381	0.2955712	66

(b) Table 10: provincial electoral results.

7 Short summary

- Sapienza studies whether government-owned banks lend differently from private banks.
- The empirical setting is Italy from 1991 to 1995, when state banks were large and politically connected.
- The dataset combines firm balance sheets, individual credit-line contracts, bank characteristics, political affiliations, and provincial election results.
- The author matches state-bank borrowers to private-bank borrowers with the same risk profile, industry, geography, ownership type, and similar size.
- State-owned banks charge lower rates than private banks to similar firms.
- The raw discount is around 23 basis points; in controlled regressions the discount is roughly 44–50 basis points.
- The result survives firm fixed effects, time fixed effects, bank controls, market concentration controls, and dual-borrower sample restrictions.
- The discount is not explained by credit constraints because it remains for firms borrowing from both types of banks and with unused private credit lines.
- State-bank discounts are larger in southern Italy and for large firms.
- The larger discount for large firms is difficult to reconcile with a simple social view of lending to credit-constrained small firms.
- The strongest evidence for political lending is that state banks charge lower rates in provinces where their affiliated political party is stronger.
- The paper concludes that state-owned banks can function as a mechanism of political patronage.

8 Expanded conclusion

8.1 Core conclusion

The paper's central conclusion is that government bank ownership affects the pricing of credit. State-owned banks do not merely look different in aggregate profitability or balance-sheet composition; they charge systematically lower interest rates to firms that are comparable to, or identical to, firms borrowing from private banks. The most persuasive estimate is roughly a 44–50 basis point discount after controlling for firm, bank, time, and market characteristics.

8.2 Why the evidence favors the political view

The social, agency, and political views can each explain part of the evidence, but only the political view naturally explains all of it. A social view can explain cheaper loans in poorer southern regions, but it struggles to explain why large firms benefit more than small firms and why discounts are connected to affiliated parties' local vote shares. The agency view can explain inefficient or overly generous lending, but it does not predict that discounts should rise in provinces where the bank chairperson's political party is stronger. The political view predicts targeted transfers to electorally valuable borrowers and locations, which is exactly the pattern in the final tests.

8.3 Policy implication: public banking needs governance, not only public objectives

The paper does not imply that every public bank is harmful or that all state credit programs are politically captured. It does imply that public ownership alone is not enough to guarantee social lending. If state-owned banks lack insulation from politicians, subsidized credit can become a hidden transfer mechanism. Good public banking therefore requires governance safeguards: transparent lending criteria, independent loan committees, public reporting, restrictions on political appointments, and ex post audits of who receives subsidized credit.

8.4 Financial-development implication

Politicized credit allocation can weaken financial development through several channels. First, it reduces the information content of interest rates. Second, it reallocates bank capital toward politically connected borrowers rather than high-productivity borrowers. Third, it may crowd out private monitoring because borrowers can seek political discounts rather than improve credit quality. Fourth, if subsidized loans are repeatedly rolled over, state banks may accumulate weak portfolios and require public support.

8.5 Growth and productivity implication

The allocation effect is potentially more important than the average interest-rate discount. A 44 basis point discount is economically meaningful, but the deeper issue is whether capital flows to firms with political value rather than productive value. If large politically visible firms receive favorable credit, while smaller innovative or financially constrained firms do not, state bank lending may protect incumbents and slow reallocation. This mechanism connects the paper to broader evidence that state ownership of banks can be associated with lower financial development and weaker productivity growth.

8.6 Interpretation for modern policy debates

The paper remains relevant for debates over development banks, green banks, regional credit programs, and emergency credit facilities. Public credit can be justified when externalities or credit-market failures are real. But the paper shows that the policy objective and the political implementation are separate. Even a program designed to address a real market failure can become patronage if loan terms are discretionary and political actors influence bank management.

8.7 What the paper teaches empirically

The paper's empirical strategy is a useful template for studying political allocation. Instead of comparing public and private institutions at the aggregate level, it compares contract terms at the transaction level. Instead of treating political influence as an abstract institutional feature, it maps bank leadership to political parties and parties to local vote shares. This combination of micro contracts and political data is what makes the patronage interpretation credible.

8.8 Limitations and external validity

The evidence is from Italian credit-line lending in the early 1990s. The results may not generalize one-for-one to countries where public banks have stronger independence, where political appointments are limited, or where public lending is rule based. The paper also cannot observe all loan characteristics, such as collateral quality, service quality, or the probability that a line is revoked. These caveats matter, but they do not overturn the main contribution: when political links are measurable, state-bank lending terms move with those links.

8.9 Research agenda

A natural next step is to measure the real effects of these subsidized loans: employment, investment, productivity, survival, and default. Another important question is whether political discounts are repaid through campaign support or local employment. A third question is whether privatization, governance reform, or changes in appointment rules reduce political lending. Finally, similar methods could be applied to modern public credit programs, especially those justified by regional development or climate policy.

8.10 Takeaway

The paper's enduring message is that ownership matters because objectives matter. Private banks may underprovide socially valuable credit, but state-owned banks may overprovide politically valuable credit. The welfare question is therefore not public versus private in the abstract. It is whether the governance structure of public finance can deliver social objectives while preventing political capture.